

Universidad El Bosque

Course:

Databases II

Final Project:

Payroll Management System

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1. INTRODUCTION

This document describes the design, development, and implementation of a Payroll Management Information System in Colombia, addressing the database design, system architecture, implemented functional modules, development approach, and technologies used. This system seeks to automate, optimize, and guarantee the traceability of processes related to employee management and legal payroll calculations under the Colombian regulatory framework.

A hybrid architecture is being considered, including the use of a relational database (**SQL Server**) for operational and transactional modules, such as updates, accruals, deductions, payroll, auditing, among others; and a NoSQL database (**MongoDB**) for document management, specifically for employee resumes, allowing for flexible, non-relational management of personal and employment data.

The system was developed using **Spring Boot**, which facilitated the creation of RESTful services and the management of the business layer, along with **HTML5, CSS3 and the Thymeleaf** template engine for the user interface, following an MVC (Model-View-Controller) architecture.

2. SYSTEM OBJECTIVES

- Registration and control of staff updates.
- Calculation and storage of overtime.
- Settlement of accruals and deductions.
- Generation of the monthly payroll.
- Calculation of social benefits and employer contributions.
- Settlement and control of payments corresponding to social benefits on a semi-annual basis.
- Generation of payroll slips in PDF.
- Historical audit of payroll processes.
- Document management of resumes in a NoSQL database.

3. TECHNOLOGIES USED

- **Spring Boot:** Main framework for backend development, facilitating the creation of REST APIs and integration with services.
- **SQL Server:** Relational database where all payroll transactional information is stored and managed.
- **MongoDB Atlas:** Document database used to store employee resumes, allowing flexibility and scalability.
- **Thymeleaf:** Template engine used to integrate dynamic backend data into HTML views.
- **CSS3 and HTML5:** Base technologies for the design and visual structure of the application, ensuring compatibility with different browsers and accessibility for users.
- **JavaScript and SweetAlert2:** Form validation, DOM manipulation, and interactive alerts for a better user experience.

4. GENERAL SYSTEM ARCHITECTURE

The system was designed with a layered architecture that separates responsibilities and makes it easy to maintain:

- **Presentation Layer:** Includes CSS-styled interactive HTML views, controlled with Thymeleaf.
- **Business logic layer:** Contains the services implemented in Java using Spring Boot, where the system rules are concentrated.
- **Persistence Layer:** Manages communication with SQL Server using JPA/ Hibernate and with MongoDB using Spring Data.
- **Data Layer:** Integrates a relational model (SQL Server) and a NoSQL model (MongoDB) to efficiently manage structured and unstructured data.

5. RELATIONAL DATABASE MODEL (SQL SERVER)

The payroll management system database is fully normalized to Third Normal Form (3NF), ensuring data integrity, elimination of redundancies, and relational consistency. It is designed to support the complete payroll process flow in a Colombian company, in compliance with current labor regulations. The main tables and key stored procedures used in the system are described below.

Main tables and their purpose:

- **empleado**
Stores the personal, contractual, and administrative data of each employee. It includes fields such as document type and number, first name, last name, position, department, base salary, contract type, affiliations (EPS, ARL, pension fund), status, etc.
- **novedad**
Records extraordinary events associated with the employee that may affect their payroll (vacations, sick leave, licenses, permits, etc.). It is linked to the type of novelty and impacts the accrual calculation.
- **horas_extras**
Contains records of overtime hours worked by each employee, indicating the start and end date, the type of overtime, the number of hours, and the total paid.
- **devengado**
Summarizes the monthly accrued amounts for each employee. Includes salary, transportation allowance, total overtime, subtotal, and total accrued. Serves as a base input for calculating deductions and payroll.
- **deducciones**
Record the deductions associated with each accrual, including health, pension, solidarity fund, and withholding tax.
- **prestaciones_sociales**
Stores the calculated monthly values of the four mandatory social benefits: severance

pay, severance interest, service bonus, and vacation pay. This is the basis for the semiannual settlement of benefits.

- **aportes_patronales**

Records the amounts that the company contributes monthly to social security: employer health, employer pension, ARL, and payroll taxes (Caja de Compensación, SENA, ICBF). It is calculated based on the employee's salary and risk class.

- **nomina**

Consolidated table that represents the monthly payroll per employee, integrating all components: accrued, deductions, contributions, benefits, and total net payable.

- **auditoria_nomina**

(via trigger) each time a row is inserted into the payroll table, storing: employee, period dates, total paid to the worker, payroll total, transaction date, and user. Provides process traceability.

- **liquidacion_prestacion**

Records the semiannual settlement of employee benefits, consolidating the value of severance pay, interest, bonuses, and accrued vacation time from the previous six months. It is used as a basis for payment or historical review.

6. DESCRIPTION OF IMPLEMENTED MODULES

The payroll management system is composed of multiple functional modules, each designed to fulfill a specific responsibility within the employee's work accounting cycle. The implemented modules, their functions, and the stored procedures associated with each are detailed below:

6.1. Employee Module

This module manages employees' personal and contractual information. It includes data such as first and last names, document type and number, base salary, start date, date of birth, marital status, EPS, ARL, pension fund, bank, contract type, and current status (active/inactive).

- **Associated tables:** empleado, cargo, departamento, eps, arl, fondo_pension, tipo_contrato, estado.
- **Relationships:** Each employee is related to multiple auxiliary entities such as title, department, and status.



6.2. Novelty Module

It allows you to record exceptional situations in an employee's workday, such as leaves of absence, sick leave, vacation, and permits. These affect payroll settlement.

- **Table:** novedad
- **SP:**
 - `sp_registrar_novedad`: Inserts a novelty with dates, type, and observations. Validates integrity and dates.
- **Key logic:** The novelties can deduct the days worked in the accrual calculation.



6.3. Overtime Module

Manages the registration and settlement of hours worked outside of conventional hours, broken down by type (daytime, nighttime, Sunday, etc.).

- **Table:** horas_extras
- **SP:**
 - sp_registrar_horas_extras_empleado: Automatically calculates the value paid per overtime hour and records it in the table.
- **Relationship:** The total paid is used as input for the accrual module.

The screenshot shows a web browser window with the address bar displaying 'localhost:8080/horas-extras'. The page title is 'Gestión de Horas Extras'. The main header is 'Sistema de Gestión de Nómina' with a navigation menu containing: Inicio, Empleados, Novedades, Horas Extras, Nómina, Liquidación de Prestaciones, Reportes, Hojas de Vida, and Administración. The main content area is titled 'Registrar Horas Extras' and contains a form with the following fields:

- Empleado:** A dropdown menu with the placeholder text 'Seleccione un empleado'.
- Tipo de Hora Extra:** A dropdown menu with the placeholder text 'Seleccione un tipo'.
- Cantidad de Horas:** A text input field.
- Fecha Inicio:** A date picker field showing 'dd/mm/aaaa'.
- Fecha Fin:** A date picker field showing 'dd/mm/aaaa'.

6.4. Accrual Module

Calculates the base salary proportional to transportation assistance and overtime pay. It represents the employee's gross total.

- **Table:** devengado
- **SP:**
 - sp_generar_devengados_mensual: Calculates the subtotal, transportation assistance, total overtime hours, and total accrued per employee. Use auxiliary functions:
 - fn_calcular_sueldo_proporcional
 - fn_calcular_auxilio_proporcional
 - fn_total_horas_extras_pagadas
- **Importance:** This module is the basis of the worker's gross payment.

Reporte de Devengados

localhost:8080/devengado

Sistema de Gestión de Nómina

Inicio Empleados Novedades Horas Extras Nómina Liquidación de Prestaciones Reportes Hojas de Vida Administración

Reporte de Devengados

Desde: 01/05/2025 Hasta: 31/05/2025 Empleado: Nombre o apellido Filtrar

ID	Empleado	Sueldo	Horas Trabajadas	Auxilio	Total Horas Extras	Total Devengado	Inicio	Fin
1	Juan Luis Vargas Soto	1,429,803.50	230.00	200,000.00	0.00	1,629,803.50	2025-05-01	2025-05-31
2	Carlos Jorge Ortega Suárez	6,806,542.25	230.00	0.00	0.00	6,806,542.25	2025-05-01	2025-05-31
3	Camila Laura Pineda Jiménez	3,760,614.47	230.00	0.00	0.00	3,760,614.47	2025-05-01	2025-05-31
4	Mateo Miguel Vargas López	1,216,170.95	230.00	200,000.00	0.00	1,416,170.95	2025-05-01	2025-05-31
5	María Andrea Moreno Suárez	5,841,152.76	230.00	0.00	0.00	5,841,152.76	2025-05-01	2025-05-31

6.5. Deductions Module

Manages applicable deductions from employee salaries, such as health, pension, and withholding taxes, among others.

- **Table:** deducciones
- **SP:**
 - `sp_generar_deducciones_manual`: Calculates mandatory and other custom discounts.
- **Relationship:** Each deduction is associated with an `id_devengado`.

Reporte de Deducciones

localhost:8080/deduccion

Sistema de Gestión de Nómina

Inicio Empleados Novedades Horas Extras Nómina Liquidación de Prestaciones Reportes Hojas de Vida Administración

Reporte de Deducciones

Desde: 01/05/2025 Hasta: 31/05/2025 Empleado: Nombre o apellido Filtrar

ID	Empleado	Salud	Pensión	Fondo Solidaridad	ReteFuente	Días Descontados	Total	Inicio	Fin
1	Juan Luis Vargas Soto	57,192.14	57,192.14	0.00	0.00	0.00	114,384.28	2025-05-01	2025-05-31
2	Carlos Jorge Ortega Suárez	272,261.69	272,261.69	68,065.42	0.00	0.00	612,588.80	2025-05-01	2025-05-31
3	Camila Laura Pineda Jiménez	150,424.58	150,424.58	0.00	0.00	0.00	300,849.16	2025-05-01	2025-05-31
	Mateo Miguel Vargas López								

6.6. Social Benefits Module

Calculate monthly legal benefits such as severance pay, severance interest, service bonuses, and vacation pay.

- **Table:** prestaciones_sociales
- **SP:**
 - sp_generar_prestaciones_sociales_mensual: Inserts benefits accrued per month, for each active employee.
- **Relationship:** Cumulative data that is then used in the semi-annual settlement.



ID	Empleado	Cesantías	Intereses	Prima	Vacaciones	Total Prestaciones	Inicio	Fin
1	Juan Luis Vargas Soto	135,762.63	16,291.52	135,762.63	59,574.19	347,390.97	2025-05-01	2025-05-31
2	Carlos Jorge Ortega Suárez	566,984.97	68,038.20	566,984.97	283,601.39	1,485,609.53	2025-05-01	2025-05-31
3	Camila Laura Pineda Jiménez	313,259.19	37,591.10	313,259.19	156,689.76	820,799.24	2025-05-01	2025-05-31
4	Mateo Miguel Vargas López	117,967.04	14,156.04	117,967.04	50,672.98	300,763.10	2025-05-01	2025-05-31
5	María Andrea Moreno Suárez	486,568.02	58,388.16	486,568.02	243,377.47	1,274,901.67	2025-05-01	2025-05-31

6.7. Employer Contributions Module

Calculate the contributions that the employer must make to social security (health, pension, occupational risks, and parafiscal contributions).

- **Table:** aportes_patronales
- **SP:**
 - sp_generar_aportes_patronales_mensual: Calculates the total value of contributions per employee and period.
- **Impact:** It is part of the total payroll cost for the company.

Reporte de Aportes Patronales

localhost:8080/aporte-patronal

Sistema de Gestión de Nómina

Inicio Empleados Novedades Horas Extras Nómina Liquidación de Prestaciones Reportes Hojas de Vida Administración

Reporte de Aportes Patronales

Desde: 01/05/2025 Hasta: 31/05/2025 Empleado: Nombre o apellido Filtrar

ID	Empleado	Salud	Pensión	ARL	Caja	ICBF	SENA	Total	Inicio
1	Juan Luis Vargas Soto	121,533.30	171,576.42	34,887.21	57,192.14	42,894.11	28,596.07	456,679.25	2025-05-01
2	Carlos Jorge Ortega Suárez	578,556.09	816,785.07	35,394.02	272,261.69	204,196.27	136,130.85	2,043,323.99	2025-05-01
3	Camila Laura Pineda Jiménez	319,652.23	451,273.74	261,738.77	150,424.58	112,818.43	75,212.29	1,371,120.04	2025-05-01
4	Mateo Miguel	103,374.53	145,940.51	52,903.44	48,646.84	36,485.13	24,323.42	411,673.87	2025-05-01

6.8. Payroll Module

It functions as the core of the system, integrating accruals, deductions, social benefits, and contributions into a single monthly record for each employee.

- **Table:** nomina
- **SP:**
 - sp_generar_nomina_mensual: Checks for duplicates, generates all components per employee and consolidates the result.
- **Key fields:** total_a_pagar_empleado, total_nomina.

Generar Nómina

localhost:8080/nomina-mensual

Sistema de Gestión de Nómina

Inicio Empleados Novedades Horas Extras Nómina Liquidación de Prestaciones Reportes Hojas de Vida Administración

Generar Nómina Mensual

Fecha Inicio: 01/05/2025 Fecha Fin: 31/05/2025 Generar Nómina

Desde: 01/05/2025 Hasta: 31/05/2025 Empleado: Nombre o apellido Filtrar Nóminas

ID	Empleado	Fecha Inicio	Fecha Fin	Total Devengado	Total Deducciones	Prestaciones	Aportes	Total a Pagar	Total Nómina
1	Juan Luis Vargas Soto	2025-05-01	2025-05-31	1,629,803.50	114,384.28	347,390.97	456,679.25	1,515,419.22	2,433
2	Carlos Jorge Ortega Suárez	2025-05-01	2025-05-31	6,806,542.25	612,588.80	1,485,609.53	2,043,323.99	6,193,953.45	10,33
3	Camila Laura Pineda Jiménez	2025-05-01	2025-05-31	3,760,614.47	300,849.16	820,799.24	1,371,120.04	3,459,765.31	5,952

6.9. Reports and Detachables Module

It allows you to generate individual and mass reports in PDF format for each employee, with a breakdown of income, deductions, and net worth.

- **PDF template:** desprendible_pdf.html
- **Backend logic:** Use Thymeleaf + openhtmltopdf to render and export in ZIP.
- **Indirectly related SPs:** Requires that the previous modules have correctly generated their data.



6.10. Audit Module

Records each entry in the payroll table to ensure process traceability and control of the user who executed it.

- **Table:** auditoria_nomina
- **Trigger:**
 - trg_auditar_nomina_insercion: It is executed automatically when inserting into payroll.
- **SP:**
 - sp_buscar_auditoria_nomina_por_usuario: Allows you to filter audits by user and date.

Sistema de Gestión de Nómina

Auditoría de Nómina

Desde: 01/05/2025 Hasta: 31/05/2025 Usuario: Nombre de usuario Filtrar

ID Auditoría	Empleado	Periodo	Total Pagado	Total Nómina	Fecha de Operación	Usuario
1	Juan Luis Vargas Soto	01/06/2025 a 30/06/2025	1,515,419.22	2,433,873.72	12/05/2025 02:45	sa
2	Carlos Jorge Ortega Suárez	01/06/2025 a 30/06/2025	6,193,953.45	10,335,475.77	12/05/2025 02:45	sa
3	Camila Laura Pineda Jiménez	01/06/2025 a 30/06/2025	3,459,765.31	5,952,533.75	12/05/2025 02:45	sa
4	Mateo Miguel Vargas López	01/06/2025 a 30/06/2025	1,318,877.27	2,128,607.92	12/05/2025 02:45	sa
5	María Andrea Moreno Suárez	01/06/2025 a 30/06/2025	5,315,449.01	8,899,942.48	12/05/2025 02:45	sa

6.11. Benefit Settlement Module (Semi-annual)

It consolidates the data generated month by month in social benefits and records them definitively for each semester.

- **Table:** liquidacion_prestacion
- **SPs:**
 - sp_generar_liquidacion_prestaciones_semestre: Calculates and records settlement per semester (1 or 2).
 - sp_buscar_liquidacion_prestaciones_por_nombre: Allows you to search by year, semester and name.
- **Purpose:** Formalizes the accumulated value of benefits per period.

Sistema de Gestión de Nómina

Liquidación de Prestaciones Sociales

Año: 2025 Semestre: Enero - Junio Generar Liquidaciones

Año: 2025 Semestre: Enero - Junio Empleado: Nombre o apellido Buscar

ID	Empleado	Año	Semestre	Cesantías	Intereses	Prima	Vacaciones	Total	Generado
1	Juan Luis Vargas Soto	2025	1	271,525.26	32,583.04	271,525.26	119,148.38	694,781.94	12/05/2025 18:09
2	Carlos Jorge Ortega Suárez	2025	1	1,133,969.94	136,076.40	1,133,969.94	567,202.78	2,971,219.06	12/05/2025 18:09
3	Camila Laura Pineda Jiménez	2025	1	626,518.38	75,182.20	626,518.38	313,379.52	1,641,598.48	12/05/2025 18:09

7. DOCUMENTARY MODULE WITH MONGODB (RESUMES)

This module allows you to register, search, and delete resumes in MongoDB. Each document stores:

- Personal information: document type and number, first name, last name, telephone number, address, email, date of birth.
- Academic information: educational level, studies, certifications, skills.
- Work experience: list of items with company, position, start/end date, description.

MongoDB allows you to work without a fixed schema, facilitating the management of nested lists and optional fields, making this type of variable and documentary information ideal.



The screenshot displays a web browser window with the address bar showing 'localhost:8080/hojas-de-vida'. The application's header is dark blue with the title 'Sistema de Gestión de Nómina' and a navigation menu with links: Inicio, Empleados, Novedades, Horas Extras, Nómina, Liquidación de Prestaciones, Reportes, Hojas de Vida, and Administración. The main content area is titled 'Hojas de Vida' and features a white registration form titled 'Registrar Hoja de Vida'. The form contains the following fields: 'Tipo de Documento:' with a dropdown menu showing 'Cédula de Ciudadanía', 'Número de Documento:', 'Primer Nombre:', 'Segundo Nombre:', and 'Primer Apellido:'.

8. CONCLUSIONS

The development of the payroll management information system in Colombia has enabled the consolidation of a robust, reliable environment aligned with the country's current legal provisions. Through a Spring Boot- based architecture, the combined use of relational (SQL Server) and NoSQL (MongoDB) databases, and presentation technologies such as HTML, CSS, and Thymeleaf, a comprehensive platform was built capable of accurately and efficiently managing each of the processes associated with the payment of salaries and employee benefits.

During implementation, Colombian labor regulations established in the Substantive Labor Code and other complementary regulations were respected. The system correctly calculates:

- **Monthly accruals**, including proportional salary, transportation assistance and overtime, in accordance with the established legal limits and factors.

- **Mandatory deductions**, such as health, pension and withholding tax, according to the percentages stipulated by law.
- **Social benefits** (severance pay, interest, service bonuses, and vacations), both monthly and semiannually, ensuring compliance with legal deadlines for their payment.
- **Employer contributions**, which include employer contributions to the social security system and to parafiscal entities.
- **Automated auditing** of payroll records, providing traceability, transparency, and control over who generated the information and when.

Additionally, a complementary module was incorporated into MongoDB for resume management, demonstrating the system's ability to integrate multiple technologies based on data type and specific needs. This module allows for flexible storage of employees' professional, educational, and employment information .

The system not only meets the technical and functional requirements outlined at the start of the project, but also strictly adheres to the regulatory guidelines of the Colombian labor market. This makes it a valuable tool for both the human talent department and the accounting and administrative processes of organizations, ensuring accurate, automated, and legally valid payroll management.